

29 D

$$p = \sqrt{2mE_k}$$

$$\frac{\Delta p}{p} = \frac{1}{2} \frac{\Delta E_k}{E_k} = \frac{1}{2} (0.0010) = 0.00050$$

$$\Delta p = 0.00050p$$

$$\Delta x \geq \frac{h}{\Delta p} = \frac{h}{0.00050p} = 2000 \frac{h}{p}$$

20 B If a red filter is used, only red light incidents on the metal. Red light photons have lower